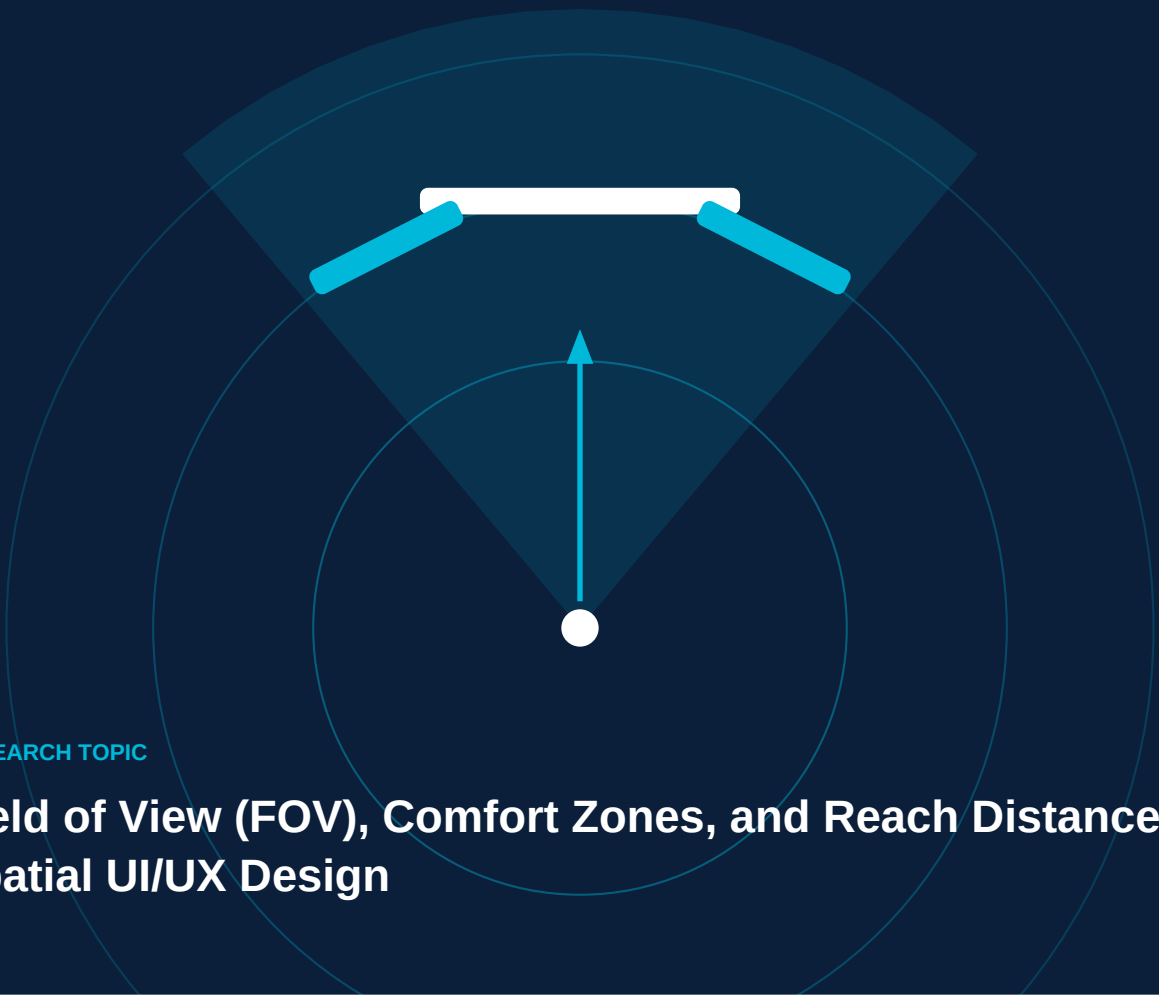
UX RESEARCH REPORT

OmniSpace

Spatial AR Operating System Design System



RESEARCH TOPIC

Field of View (FOV), Comfort Zones, and Reach Distance in Spatial UI/UX Design

MEMBER ROLE	Member 1 — UX Research and Spatial Ergonomics
PROGRAMME	UI/UX Design Internship · Infotact Solutions
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Key findings at a glance

- Main content belongs near the centre of the view; edges suit secondary actions [1][2].
- Neck turns more easily left–right than up–down, so favour wide, horizontal layouts [3].
- Keep interactive content at one depth; frequent focus changes cause eye strain [2][4].
- Look + pinch from a relaxed arm is the primary comfortable input; raised arms cause fatigue [2][4][6].
- Minimum target area is 60 points (Apple's unit), built from size plus spacing [2][3].
- Numeric angles and distances in this report are HoloLens values [4] and must be treated as provisional.

How the boxes in this report are used

VERIFIED FINDING	States what a cited source says.
DESIGN RECOMMENDATION	My own proposal for OmniSpace; not a source claim.
LIMITATION	What the evidence does not prove.
NOTE	Reading guidance or terminology.

1. Abstract and Introduction

1.1 Abstract

This report studies three ergonomic questions that every spatial interface must answer: **where** content should appear in the user's view (field of view and comfort zones), **how far away** it should be, and **how the user should reach or select it** (reach distance, gaze and pinch). The evidence comes from three Apple WWDC23 spatial-design sessions [1][2][3], two industry comfort guidelines from Microsoft and Meta [4][5], and two peer-reviewed human-computer interaction (HCI) papers on arm fatigue and gaze-plus-pinch input [6][7].

The main result is a consistent pattern across sources: keep the main content near the centre of the view, close to a neutral head position; keep interactive content at one consistent depth; prefer looking and pinching from a relaxed posture over holding the arms up; and treat direct touch as a special case. Numerical limits exist only in some sources (mostly Microsoft's HoloLens guidance [4]) and are specific to those devices. The report therefore separates **verified findings** from **design recommendations** for OmniSpace, and lists what the team still has to test on its own target hardware.

1.2 Introduction

OmniSpace is a design system for a spatial augmented-reality (AR) operating system. A **design system** is a shared set of rules, components and values (colours, spacing, sizes, behaviours) that keeps every screen consistent. On a phone, those rules are mostly about pixels. In spatial computing the user wears a headset, so the interface is placed *around* the person, and the human body — eyes, neck, shoulders, arms — becomes part of the design problem. This is called **ergonomics**: fitting the design to the body so that it can be used comfortably for a long time.

My role in the team is Member 1 — UX Research and Spatial Ergonomics. My job is to give the team a reliable, evidence-based starting point for *where* interface elements may be placed and *how far* users must move to use them, before layouts and components are finalised.

> NOTE
>
> **How to read this report**
>
> Every statement that comes from a source carries a citation number such as [4]. Boxes labelled **Verified finding** summarise what a source states. Boxes labelled **Design recommendation** are my own proposals for OmniSpace and are **not** claims made by the sources. Values marked **provisional** must be tested before they become design-system rules.

2. Research Objectives and Scope

2.1 Objectives

1. Explain field of view (FOV) and what it means for placing interface elements.
2. Define spatial comfort zones and identify where primary and secondary interface elements should and should not go.
3. Describe how reach distance, arm fatigue, gaze-based selection and pinch gestures affect interaction comfort.

4. Produce a labelled spatial layout diagram that the whole team can use as a shared reference.
5. Translate the evidence into practical, clearly labelled recommendations for the OmniSpace design system.
6. State honestly what the evidence does *not* prove and what needs user testing.

2.2 Scope

This is a desk-research study. It reviews published guidelines and research; it does **not** include new experiments with users or hardware. It covers head-mounted spatial interfaces operated with eyes and hands (and, where relevant, voice). It does not cover visual styling (colour, materials, iconography), 3D content design, audio, or motion-sickness and locomotion design, except where these directly affect comfort and placement.

In scope	Out of scope
FOV, comfort zones, viewing distance	Brand colours, typography choices, icon design
Head and neck comfort, text readability	3D model and environment design
Reach, arm fatigue, gaze selection, pinch	Locomotion and motion-sickness design
Target size and spacing for comfortable selection	Voice-assistant and audio design

LIMITATION

Important assumption

The OmniSpace target headset was not specified in the brief I worked from. All recommendations are therefore device-neutral, and every number is a **starting value** to be replaced by measurements on the chosen hardware.

3. Field of View (FOV)

3.1 Definition

Field of view (FOV) is the part of the world a person can see without moving their head. Meta's design guidance describes it exactly this way and adds that a headset's display FOV is smaller than a person's natural FOV [5]. Apple makes the same distinction for its platform: the canvas for apps is effectively unlimited, but people only see the content that is inside their field of view [2].

3.2 Why FOV matters

In a normal phone or laptop screen, the designer knows exactly where the screen edges are. In spatial design the "screen" is wherever the user is looking, and the user can turn around. If important content sits outside the FOV, the user must turn their head or body to find it, which costs time and effort. Two sources explain what happens inside the FOV:

- **The centre is easiest.** Apple states that it is easiest to see things in the centre of the view, so the most important content should go there [1]. Apple's input session adds that looking at the edges of the FOV can be tiring for the eyes, so edge areas suit content that is not needed all the time, such as secondary actions [2].

- **Wide, not tall.** Apple notes that the FOV is wide, so landscape layouts suit it [1]. Apple's UI session adds that, because of the neck's range of motion, it is easier for most people to turn the head left and right than up and down, and recommends being careful with content placed too high or too low [3].
- **Do not glue content to the view.** Apple advises against anchoring content to the person's view because it feels stuck and can be disorienting; content should instead be anchored in the person's space [1]. Meta gives related advice: content fixed in front of the camera can hide a large part of the real world and reduces awareness of surroundings [5]. Microsoft recommends body-locking over head-locking for heads-up displays, because displays rigidly locked to head orientation are likely to cause discomfort [4].
- **Main content should fit inside the FOV.** People do look around, so immersive content can extend beyond the FOV, but Apple says main content should generally stay inside it, otherwise it is hard to read or understand all at once [1].

3.3 Viewing comfort: depth and focus

Comfort is not only about *where* content is in the view but also *how far away* it is. Apple explains that our eyes focus on one distance at a time, and changing focus depth frequently can create eye strain, so interactive content should stay at the same depth [2]. Microsoft explains the underlying cause, the **vergence–accommodation conflict**. In real life, the eyes turn inwards (*vergence*) and adjust their lens focus (*accommodation*) together. On most headsets the display is focused at a fixed distance while the eyes turn to the virtual object's distance, so the two cues disagree, which can cause visual discomfort or fatigue [4].

VERIFIED FINDING

Verified finding — viewing distance (HoloLens-specific numbers)

For HoloLens devices, Microsoft states that the optimal zone for hologram placement is between 1.25 m and 5 m, that designers should encourage interaction from 1 m or farther, and that content should fade out at 40 cm with a clipping plane at 30 cm [4]. Microsoft also states that the display's optical distance is about 2.0 m, and that content moving in depth or forcing rapid near/far focus changes is more likely to cause discomfort [4]. **These values belong to HoloLens; they are not universal.** Apple's cited sessions give no distance in metres and say only to place content a bit further than arm's reach in most cases [1].

3.4 Implications for spatial interface placement

Finding (source)	What it means for placement
Centre of FOV is easiest to see; edges tire the eyes [1][2]	Put the main task content near the centre. Use the edges for secondary actions only.
FOV is wide; neck turns more easily left–right than up–down [1][3]	Spread layouts horizontally. Avoid stacking important content high or low.
Eyes focus on one depth at a time [2]	Keep interactive panels at a similar distance. Open pop-ups at the same distance as their parent.
Anchoring to the view feels stuck [1]; head-locked displays cause discomfort [4]	Anchor panels in the room or to the body, not rigidly to the head.
Headset FOV is smaller than natural vision [5]	Test on the real device; do not design from assumptions about human vision alone.

4. Spatial Comfort Zones

4.1 Definition

A **comfort zone** is a region around the user in which content can be seen and used without strain. Microsoft uses the term "zone of comfort" for viewing distance [4], and describes optimal and avoided ranges for gaze direction [4]. Apple describes the same idea in plain language: place content comfortably in all dimensions along a natural line of sight [1].

> NOTE
>
> **A note on terminology**
>
> The three-tier labels used in this report (**primary**, **secondary**, **avoid**) are a working framework I built by combining the sources. They are not official terms from Apple, Microsoft or Meta.

4.2 Head and neck comfort

Microsoft gives the most specific neck and gaze numbers found in this research. They were written for HoloLens and Windows Mixed Reality, so they are best treated as reference points, not laws [4]:

Direction	Microsoft guidance [4]	Plain meaning
Up	Avoid gaze angles more than 10° above the horizon	Do not make people look up for normal tasks.
Down	Avoid gaze angles more than 60° below the horizon	Do not place content near the feet.
Sideways	Avoid neck rotations more than 45° off-centre	Keep panels within a modest head turn.
Resting gaze	10°–20° below the horizon is the optimal resting angle	Heads tilt slightly down by habit.
Content region	Optimal region is 0° to 35° below the horizon	Place the main content slightly below eye level.

Microsoft also advises that horizontal head movement suits frequent interactions, while vertical movement should be reserved for uncommon events [4]. Apple's guidance agrees in direction: by default, windows are placed along a natural line of sight to encourage a healthy, comfortable posture, and content should be placed relative to the person's head in the direction they are facing, which also accounts for people of different heights and for postures such as lying back on a couch [1]. Apple also advises avoiding content behind the user or extremely high or low unless it is part of an immersive experience [1].

4.3 Primary and secondary interface placement

Zone	Where (conceptual)	What goes there	Source basis
Primary	Centre of FOV; near neutral head position; slightly below eye level	The main task and the content the user looks at most	[1][2][4]
Secondary	Left and right of the primary panel, or just below it; inside a modest head turn	Less-frequent controls, tools, reference content	[2][3][4]
Avoid	Behind the user; far above; far below; far off to the sides	Nothing important, unless the experience is deliberately immersive	[1][4]

Apple's own interface components follow this idea. In its UI session, the tab bar sits on the side of a window and expands only when the user looks at it for a while, and toolbars can be placed as "ornaments" that float just in front of the window edge, so frequently used controls stay close without covering content [3]. Apple also advises keeping apps in a single window where possible, since multiple windows quickly become a lot for people to manage [1]. Meta similarly suggests keeping main controls in one UI panel instead of many floating windows [5].

4.4 Readability

Comfort zones only work if text can be read inside them. Three findings are relevant:

- **Keep text flat.** Apple says text can be distracting and hard to read when it is 3D, especially at an angle, and should stay flat [1]. Meta also advises against using depth on text components [5].
- **Face the user and keep a stable apparent size.** Apple explains that if an interface is angled it is hard to read and use, so windows turn to face the person; and that dynamic scale makes a window fill the same part of the view regardless of distance, while fixed scale makes the interface smaller as it moves away [2]. This matters for any panel that can be moved.
- **Heavier type and strong contrast.** Apple's visionOS text styles use slightly heavier weights than on iOS (for example, medium rather than regular for body text, bold rather than semibold for titles) and slightly increased tracking, and vibrancy brightens text on glass materials so it stays legible against changing backgrounds [3]. Microsoft notes that legibility depends on display properties, lens properties and font properties, and recommends testing and using font sizes as large as possible [4].

VERIFIED FINDING

Verified finding — what the sources do not give

None of the sources reviewed provides a minimum font size in millimetres or degrees that applies across devices. Microsoft says to test legibility on the specific application and display [4]. OmniSpace will need to determine its own minimum text size by testing.

5. Reach Distance and Interaction Ergonomics

5.1 Reach distance and comfortable hand interaction

Reach distance is how far the user's hands can comfortably extend. It divides spatial interaction into two modes:

- **Indirect (distance) interaction.** The user looks at a target and uses a small hand gesture while the arm stays relaxed. Apple describes this as looking at a button and tapping the fingers together while "keeping your arm relaxed on your lap", and says the system is designed for comfortable interaction at a distance [2]. Apple recommends placing content, most of the time, a bit further than arm's reach to encourage this [1]. Meta likewise recommends offering distance interaction, because direct interaction on far objects or over long periods is tiring [5].
- **Direct touch.** The user reaches out and touches virtual content with fingertips, for example to type on a virtual keyboard or manipulate a 3D object [2]. Apple lists suitable uses: close inspection, object manipulation, mechanics that build on real-world muscle memory, and physical activity [2]. Meta adds that items needing direct touch should not be placed too close to the user, as this can cause visual discomfort [5]. Microsoft says extra care is needed for any content closer than 1 m, and recommends testing reachability at near distances [4].

5.2 Arm fatigue

Holding the arms in the air tires the shoulder and upper arm, a problem HCI researchers call the **"gorilla-arm effect"**. Hincapié-Ramos et al. describe mid-air interactions as prone to fatigue and a feeling of heaviness in the upper limbs, note that designers lacked a quantitative way to measure it, and propose a metric called **Consumed Endurance** based on the biomechanics of the upper arm [6]. The three industry sources agree in practice:

- Apple states directly that holding hands in the air will cause fatigue after a while, and that direct interaction should be used only when it is central to the experience [2].
- Microsoft warns that muscle fatigue builds when users keep a hand raised, and that repeated air-tap gestures over long periods are also fatiguing; it recommends avoiding constant repeated gesture input, adding short breaks or mixing gesture with speech [4].
- Meta warns that repetitive motions can tire users and lead to physical stress [5].

> LIMITATION
>
> **Evidence limit**
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> Only the abstract and project description of the Consumed Endurance paper [6] were verified for this report. I therefore cite it for the existence of arm fatigue and a way to measure it, but I do **not** quote any specific endurance times, angles or arm positions from it.

5.3 Gaze-based selection

Apple calls eyes the primary targeting mechanism on its platform: interfaces react to where the user looks, and any element can be targeted just by looking at it, no matter how far away [2]. Apple's guidance for making elements easy to target with the eyes can be turned into concrete design-system rules:

Rule (source)	Detail
Minimum target area of 60 points [2][3]	Elements can look smaller if size plus surrounding space reaches 60 points. Example: a 44-point button with at least 8 points of empty space around it [3].
Generous spacing [2][3]	Apple suggests at least 16 points between stacked standard buttons, and about 4 points of padding between list items so hover effects do not overlap [3].
Rounded, centred shapes [2]	Circles, pills and rounded rectangles guide the eyes to the centre. Sharp edges pull attention to the outside and reduce precision [2].
Hover feedback [2][3]	Interactive elements should show a subtle highlight when looked at, so the user knows the eyes are driving the interaction.
Dynamic scale [2]	Target areas stay the same apparent size at any distance. Fixed scale makes targets too small when moved away.

LIMITATION

A unit that must be translated

A **point** in Apple's system is a scalable unit tied to its platform, which keeps apparent size roughly constant with distance [1][2]. The cited sessions do not state what visual angle 60 points equals. OmniSpace needs its own equivalent, ideally defined as an *angle* or a distance-independent unit, measured on the chosen headset.

Gaze also supports accessibility. Apple describes **Dwell Control**, where looking at a button for a short time selects it without a hand gesture [2]. Apple also notes that eye data is sensitive and that its hover effect is processed outside the app, so apps only learn what was looked at when the user triggers a gesture [2]. This privacy point is worth noting for an operating-system design system.

5.4 Pinch gestures

Apple describes pinching the fingers together as the equivalent of pressing on a phone screen, with eyes used for targeting; it also supports pinch-and-drag to scroll and two-handed zoom and rotate [2]. Because the gesture works from a resting hand position, it avoids much of the fatigue of raised arms. Apple advises relying on system-consistent gestures and defining custom gestures only when the behaviour cannot be done with the standard set. A good custom gesture must be easy to explain and perform, distinct from system gestures and everyday hand movements, repeatable without strain, low in false activations, workable with assistive technologies, and backed by a fallback UI control [2].

Peer-reviewed support for the same idea comes from Pfeuffer et al., who propose **Gaze + Pinch**: eye gaze selects targets, and indirect freehand gestures manipulate them, so the gesture can act on any object the user looks at, near or far, without extra controllers [7]. Their work is a prototype with proof-of-concept examples that were informally tested [7], so it supports the *design concept* but does not provide measured comfort limits.

VERIFIED FINDING

Verified finding — interaction summary

Across [2], [5], [6] and [7], the evidence supports this order of preference: **look + pinch from a relaxed arm** for everyday actions; **direct touch** only where close manipulation is the point, with extra feedback to compensate for the lack of touch sensation (Apple recommends visual highlight and sound cues [2]); and **alternatives** such as voice or dwell for people who cannot or prefer not to use hand gestures [2][4].

6. Spatial UI Layout Diagram

The two diagrams below combine the findings of sections 3–5 into one reference for the team. Figure 1 is a plan (top-down) view showing the user, the primary and secondary panels, the viewing direction and the interaction area. Figure 2 is a side view showing vertical comfort limits. Both are **conceptual and not to scale**.

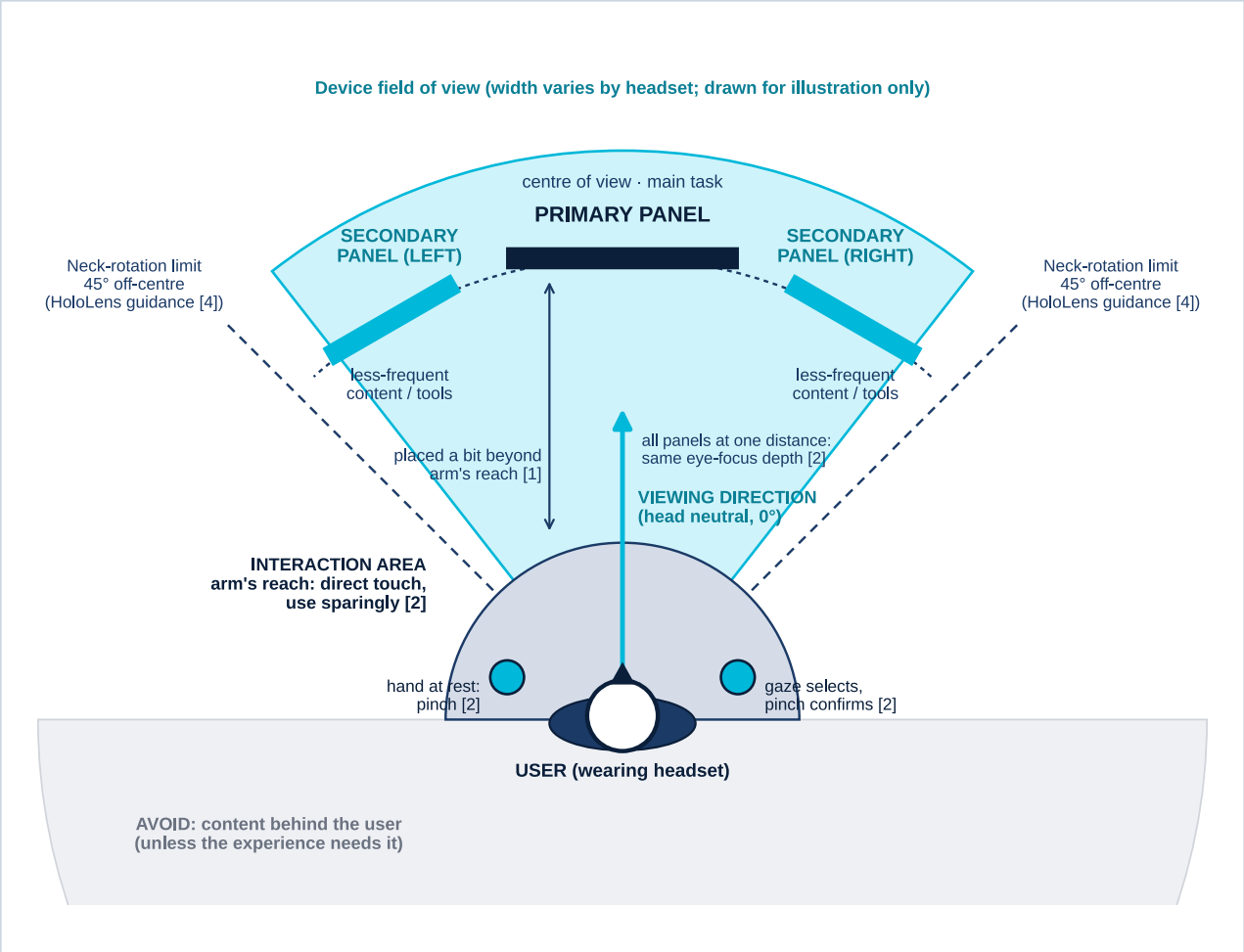


Figure 1. Top-down layout: user, primary panel, secondary panels, viewing direction and interaction area. Citation numbers refer to the reference list. The FOV width is illustrative because it differs between headsets; the 45° neck-rotation marker is Microsoft's HoloLens guidance [4].

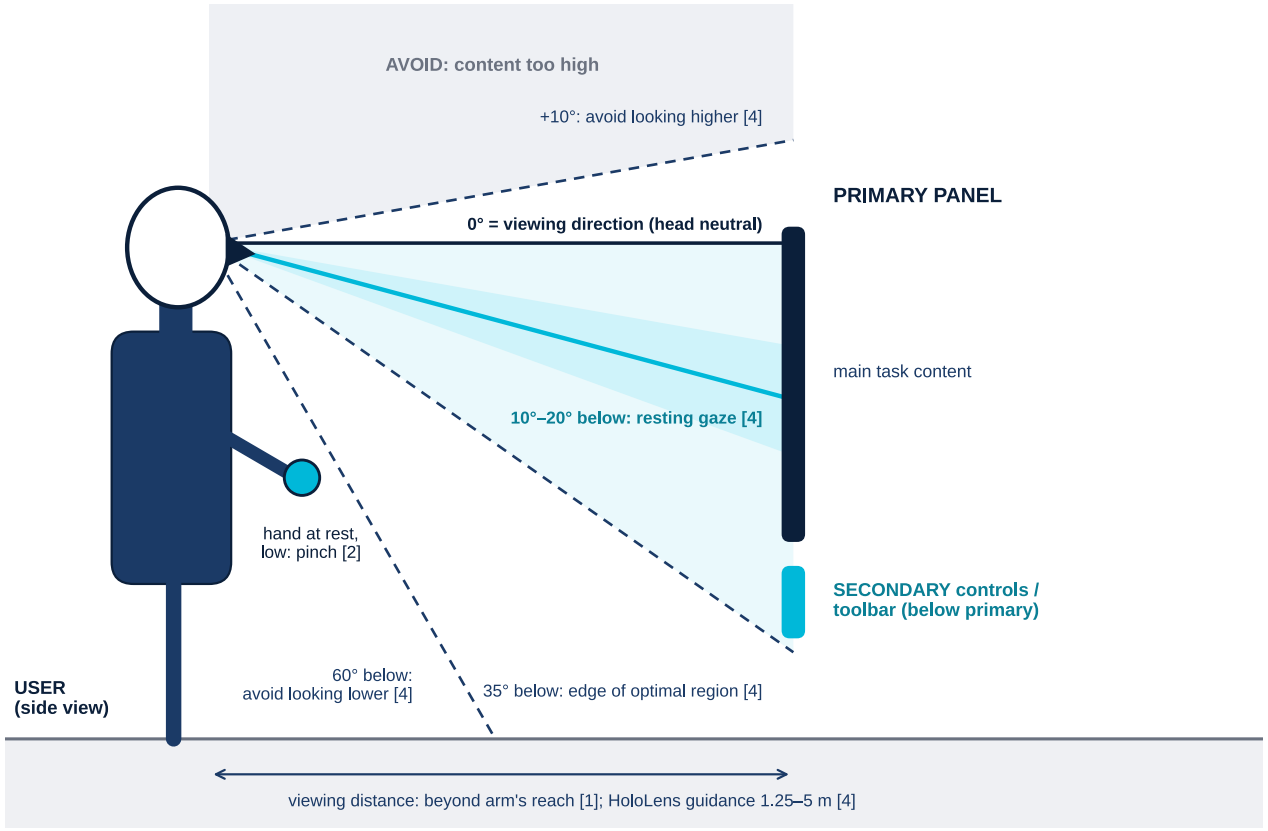


Figure 2. Side view: vertical comfort limits. All angle values (+10°, 10°–20°, 35°, 60°) are Microsoft's HoloLens / Windows Mixed Reality guidance [4]; the arm's-reach placement follows Apple [1]. Treat them as provisional for OmniSpace until tested.

How to read the diagrams

- **Primary panel** sits in the middle of the view, at a consistent distance, slightly below eye level.
- **Secondary panels** sit to the left and right (Figure 1) or just below the primary panel (Figure 2), inside a modest head turn.
- **Viewing direction** is the straight-ahead line with the head neutral; all angles are measured from it.
- **Interaction area** is the arm's-reach zone. Most selection happens at a distance (look + pinch), so this zone is reserved for deliberate direct-touch tasks.
- **Avoid** regions are behind the user, far above, and far below.

7. Research Findings and What They Mean for OmniSpace

This section links each verified finding to an implication for the OmniSpace design system. The left column is evidence; the right column is my interpretation.

#	Verified finding	Implication for OmniSpace (my interpretation)
F1	Centre of view is easiest; edges tire the eyes [1][2]	The design system needs a defined "primary zone" and "edge zone" so components know where they may appear.
F2	Neck turns more easily left–right than up–down [3]; Microsoft gives numeric gaze and neck limits for HoloLens [4]	Default multi-panel layouts should be horizontal. Vertical stacking should be rare.
F3	Eyes focus at one depth at a time; frequent depth changes cause strain [2][4]	Interactive panels should share a depth. Pop-ups and menus should open at the parent's depth.
F4	Do not anchor content rigidly to the view [1][4][5]	OmniSpace needs defined anchoring modes (world, body, head) with head-locking as the exception.
F5	Look + pinch from a relaxed arm is the primary input on Apple's platform; direct touch causes fatigue [2]	Make look + pinch the default interaction; treat direct touch as an opt-in component behaviour.
F6	A minimum of 60 points target area, built from size + spacing [2][3]	Define a minimum target size token and spacing token in a distance-independent unit.
F7	Text should be flat, face the user, and be legible; no cross-device minimum size is given [1][2][3][4][5]	Typography tokens must be tested on the final display; do not copy another platform's sizes.
F8	Arm fatigue is real and measurable [4][6]	Add a fatigue check to the team's usability testing.
F9	Alternatives such as voice and dwell exist [2][4]	Provide non-gesture paths for accessibility from the start.

8. Practical Design Recommendations for the UI/UX Team

DESIGN RECOMMENDATION

These are my recommendations, not source claims

Each recommendation names the evidence behind it. Numbers taken from a source are copied as **provisional starting values**; they must be validated on the OmniSpace headset.

8.1 Layout and placement

#	Recommendation	Evidence	Status
R1	Define three zone tokens: **Primary**, **Secondary**, **Avoid**, measured as angles from the neutral viewing direction.	[1][2][4]	Framework: mine
R2	Place the main panel centrally and slightly below eye level. Starting range: the content region 0°–35° below horizon, resting gaze 10°–20° below.	[4]	Provisional (HoloLens values)
R3	Use landscape, horizontally arranged layouts. Put secondary panels to the sides or just below, not above.	[1][3][4]	Supported
R4	Avoid placing content more than about 45° off-centre horizontally, more than 10° above, or more than 60° below the horizon.	[4]	Provisional (HoloLens values)
R5	Keep interactive panels at a similar distance. Open modals and menus at the parent's distance.	[2][4]	Supported
R6	Default placement: beyond arm's reach. Make the distance a token, with a starting range of 1.25–5 m, and fade content that comes closer than 40 cm.	[1][4]	Provisional (HoloLens values)
R7	Limit simultaneous panels. Prefer one main panel with attached secondary controls rather than many floating windows.	[1][3][5]	Supported
R8	Anchor to world or body by default; allow head-locked elements only for brief, small, essential cues. If a persistent HUD is needed, make it body-locked so it re-orients only after the head turns past a threshold.	[1][4][5]	Supported
R9	Provide a quick "recentre" action that moves open panels back in front of the user (Apple uses press-and-hold on the Digital Crown [1]).	[1]	Supported

8.2 Interaction

#	Recommendation	Evidence	Status
R10	Make look + pinch the default selection method for all standard components.	[2][7]	Supported
R11	Use direct touch only for tasks that need it (e.g., on-screen keyboard, 3D manipulation). Add proximity highlight and sound feedback to replace missing touch sensation.	[2][5]	Supported
R12	Keep sessions of repeated gestures short; offer voice or dwell alternatives and encourage breaks.	[2][4]	Supported
R13	Set a minimum target area token (starting value: 60 points in Apple's unit). Convert it to an OmniSpace unit after measuring on the headset.	[2][3]	Provisional (unit is Apple-specific)
R14	Use rounded shapes, centred labels, generous spacing, and a subtle hover highlight for every interactive element.	[2][3]	Supported
R15	Avoid custom gestures unless essential. If used, supply a visible UI alternative and test for false activations.	[2]	Supported
R16	Include seated and reclined users in all placement tests.	[1][5]	Supported

8.3 Readability

#	Recommendation	Evidence	Status
R17	Keep text flat, facing the user, and with consistent apparent size at all distances (dynamic scale or equivalent).	[1][2][5]	Supported
R18	Prefer medium or bold weights for body and title text on translucent backgrounds, and test contrast against bright and dark real-world scenes.	[3][4]	Supported (exact weights to be tested)
R19	Set minimum text size by testing on the target display, not by copying values from other platforms.	[4]	Needs testing

9. Limitations and Areas Requiring Further Testing

9.1 Limitations of this research

- **Mostly vendor guidance.** Five of the seven references are company design guidance or presentations [1][2][3][4][5], not peer-reviewed studies. They are authoritative about their own products but are not independent experiments.
- **Device-specific numbers.** All numeric angles and distances in this report come from Microsoft's HoloLens guidance [4]. Apple's sessions give point-based sizes, not angles or metres. Meta's page gives no numeric angles. No source here establishes a universal FOV, comfort angle or reach distance.

- **Limited peer-reviewed evidence.** The two research papers cover a fatigue metric [6] and a prototype gaze-plus-pinch technique [7]. Neither was used to extract numerical limits. For [6], only the abstract and project page were verified; for [7], the abstract and bibliographic data.
- **Platform differences.** Apple's sessions describe a headset that shows the real world through passthrough and tracks eyes and hands; Microsoft's guidance was written mainly for HoloLens and Windows Mixed Reality devices. The effect of these differences on comfort was not studied here.
- **No anthropometric data.** Arm length, height and neck mobility vary between people. This report does not include population data, and the sources did not provide it.
- **Video transcripts.** The three Apple sources are WWDC video sessions; I verified claims against the session transcripts and descriptions, not against a separate written specification.
- **No OmniSpace data.** No testing with OmniSpace prototypes or participants has been done.

9.2 Recommended further testing

Question to test	Suggested method	Why it matters
What are the real usable FOV and best panel angles on the OmniSpace headset?	Measure the device FOV; run placement tests at several angles with 5–8 participants	R2, R4 values are provisional
Which default distance is most comfortable?	A/B tests of 3–4 distance values; collect comfort ratings and task times	R6 value comes from HoloLens
Does look + pinch cause fatigue or errors in longer sessions?	30–45 min task sessions; rating of perceived exertion; error counts; optionally apply the Consumed Endurance approach [6]	Supports R10–R12
What is the minimum comfortable target and text size?	Target-selection tasks and reading tasks at several sizes and distances	R13, R19
Do results differ for seated, standing, reclined users and for people with different heights or mobility?	Include varied postures and participants, including accessibility needs	R16
Are timing and error rates acceptable for gaze + pinch (e.g., gaze moving away before the pinch)?	Log mis-selections in prototype tests	Prototype in [7] was only informally tested

10. Conclusion

The research shows a clear, consistent direction even though exact numbers vary by device. Spatial interfaces are most comfortable when the main content is near the centre of the view and the neutral head position, when interactive content shares one depth, when text is flat and readable, and when users select by looking and pinching with a relaxed arm instead of reaching out repeatedly. Microsoft's HoloLens guidance provides concrete starting angles and distances [4], Apple's sessions provide target-size rules and placement principles [1][2][3], Meta adds the importance of keeping content within the FOV and avoiding too many windows [5], and the HCI literature confirms that arm fatigue is real and that gaze-plus-pinch is a promising input model [6][7].

For OmniSpace, the most useful outcome of this research is a set of **zone and target tokens** with provisional values, plus a short test plan to replace those values with measurements from the team's own

hardware. The next step is to review these recommendations with the team, agree which are adopted into the design system, and run the first placement and fatigue tests.

References

Numbers in square brackets in the text match the entries below. Each URL is clickable.

[1] Apple Inc. (2023). *Principles of spatial design* [Video]. WWDC23, session 10072. Apple Developer. https://developer.apple.com/videos/play/wwdc2023/10072/

[2] Apple Inc. (2023). *Design for spatial input* [Video]. WWDC23, session 10073. Apple Developer. https://developer.apple.com/videos/play/wwdc2023/10073/

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Appendix A. Glossary of Terms

Term	Meaning
Field of view (FOV)	The part of the world visible without moving the head [5].
Headset / HMD	Head-mounted display: the device worn on the head.
Passthrough	Showing the real world through the headset's cameras. Optical see-through devices (like HoloLens) use transparent lenses instead.
Panel / window	A flat area of interface placed in space.
Head-locked / body-locked / world-locked	Content fixed to the head (moves with every head turn), to the body (follows the body, not small head turns), or to the room (stays put) [4].
Gaze	Where the eyes are looking.
Dwell	Selecting something by looking at it for a short time [2].
Pinch	Touching thumb and finger together as a click/tap gesture [2].
Direct touch	Touching virtual content with the fingertips [2].

Term	Meaning
Vergence–accommodation conflict	A mismatch between where the eyes point and where they focus, causing visual discomfort [4].
Gorilla-arm effect	Fatigue and heaviness in the arms from holding them up for mid-air interaction [6].
Point (Apple)	Apple's scalable size unit for interface elements; it keeps apparent size consistent across distances on that platform [1][2].
Ergonomics	Fitting design to the human body so use is comfortable and safe.
Design token	A named, reusable design value (such as a minimum target size).

Contribution to OmniSpace Design

Member 1 · UX Research and Spatial Ergonomics

The following are initial design recommendations derived from the desk research in this report. They are proposals for the OmniSpace team and should be checked against the target headset and user feedback.

1. Primary content

Place the main dashboard and essential information near the user?s natural forward viewing direction, subject to the selected headset?s field of view.

2. Secondary panels

Place optional tools and notifications to the sides of the primary panel so users can discover them without excessive head movement.

3. Comfortable interaction

Keep frequently used controls in an easy-to-access interaction area. Avoid designs that require users to hold their arms raised for long periods.

4. Clear visual hierarchy

Group related controls, use concise labels, and make important actions visually distinct from secondary information.

5. User flexibility

Where technically feasible, allow users to reposition or resize panels to suit their posture, viewing needs, and environment.

6. Validate before finalising

Test readability and interaction on the target headset, including seated and standing use. Treat device-specific angles and distances as provisional until validated.

Scope note: This is desk research, not a user study. The recommendations are starting points for the design team, not universal ergonomic measurements.